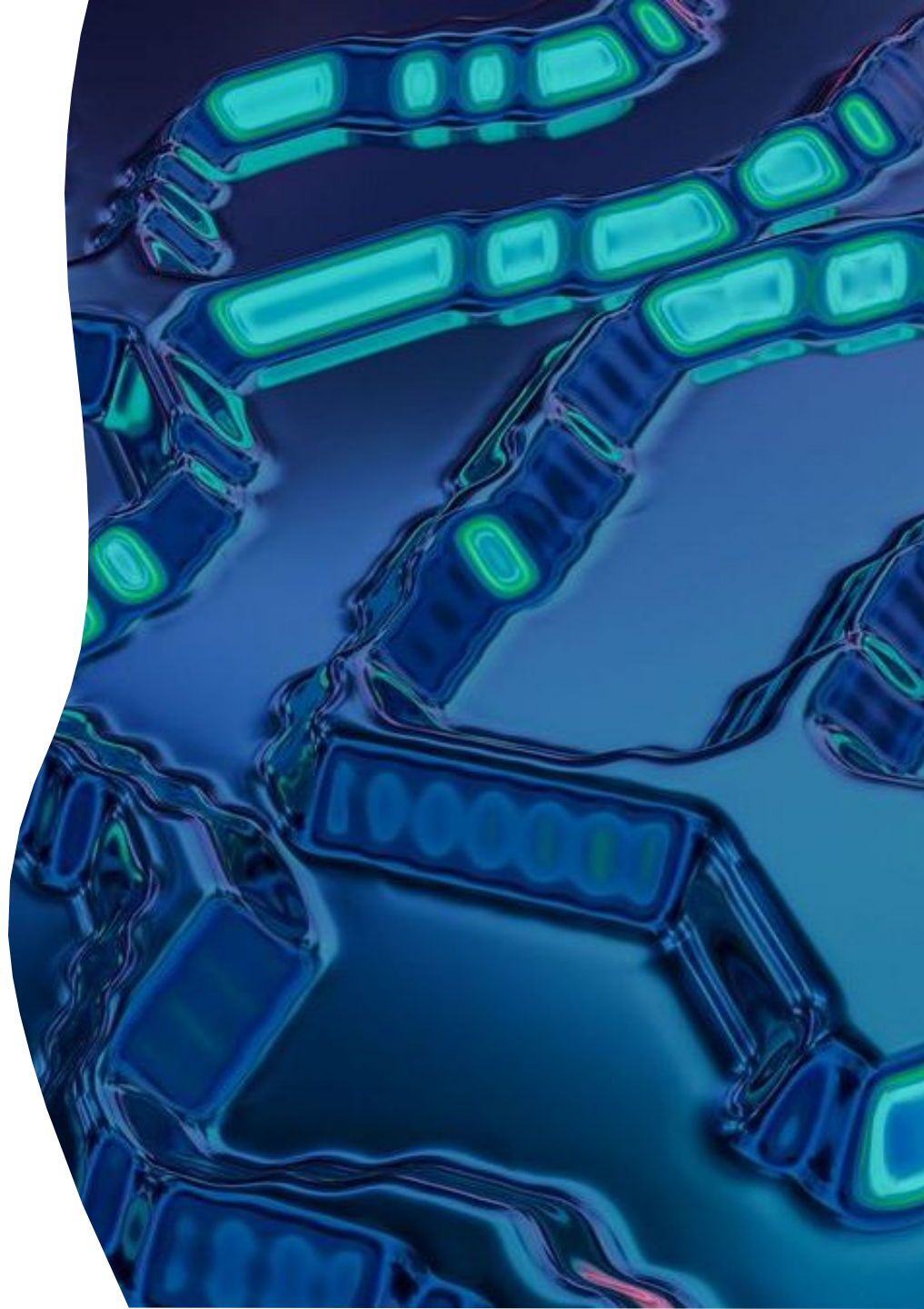
Data Analytics, Data Mining and Machine Learning

Lecture 6

Narges Chinichian · Semester 1, 2026



Overview of this course

Week	Date	Topic
1	07.09.2026	Introduction
2	14.09.2026	Data, Research Questions & Ethics
3	21.09.2026	Data Visualisation & Exploratory Data Analysis
4	28.09.2026	Preprocessing & Feature Engineering
5	05.10.2026	Machine Learning Fundamentals & Regression
6	12.10.2026	Project Presentations I
7	19.10.2026	Classification: Logistic Regression, KNN [Decision Trees is moved to the next session]
8	26.10.2026	Bank Holiday — No Class
9	02.11.2026	Gradient Descent, Regularisation & Model Complexity [+ Decision Trees]
10	09.11.2026	Model Evaluation & Ensemble Methods
11	16.11.2026	Neural Networks, Unsupervised Learning & Special Topics
12	23.11.2026	Project Presentations II

What did we learn from the feedback form?



From Linear Regression to Classification

Last teaching session: Linear Regression

Predict a **numerical value**.

Examples:

- House price → €350,000
- Energy consumption → 4,200 kWh
- Exam score → 73

Today: Classification

Predict a **category**.

Examples:

- Pass / Fail (for exams)
- Spam / Not spam (for emails)
- Fraud / Legitimate (from calls)



What Is a Classification Problem?

We have:

Features (X)

information about each observation

Target (y)

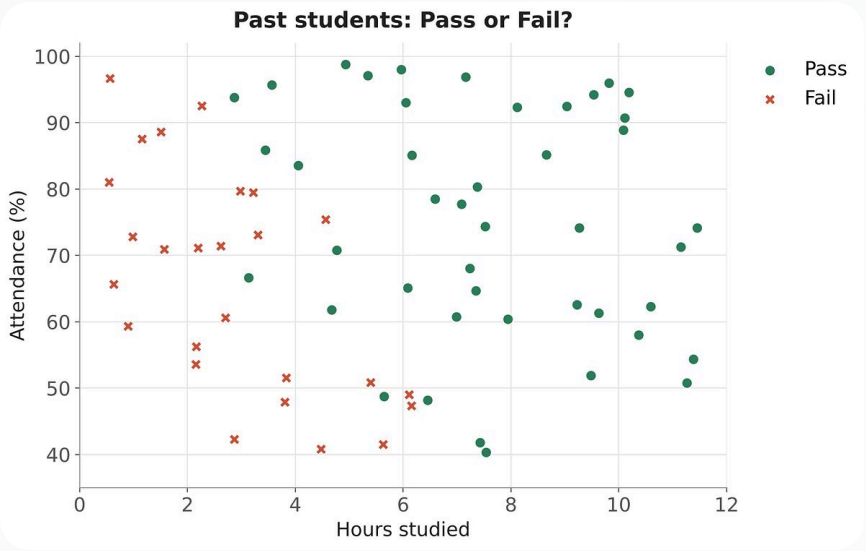
the category we want to predict

For binary classification:

$$y \in \{0, 1\}$$

We label the class of interest (the positive class) as 1.

Today: We focus on binary classification to make our models and visualisation simpler.



A Simple Example

Can we predict whether a student passes?

Features

- Hours studied
- Attendance

Target

- Pass → $y = 1$
- Fail → $y = 0$

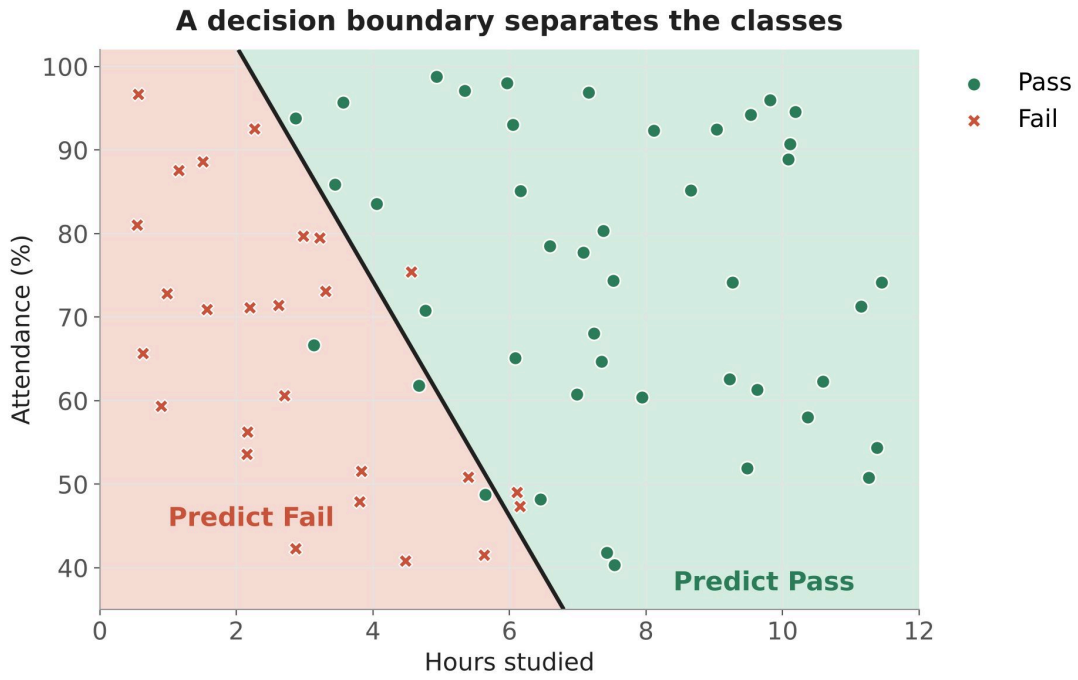
Question: Can we learn a rule from past students that predicts the outcome of a new student?

How Can We Separate the Classes?

Imagine the observations plotted in feature space.

We want to determine: which class does a new observation belong to?

One way is to learn a **decision boundary**.



Logistic Regression

The idea

Start with a linear model:

$$z = b_0 + b_1 x_1 + \cdots + b_p x_p$$

z is the **log-odds** of the positive class. It can take any value from $-\infty$ to $+\infty$.

But we want a probability between 0 and 1.

So we transform the linear score.

The Sigmoid Function

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

The sigmoid maps any value to:

$$0 < \sigma(z) < 1$$

Very negative

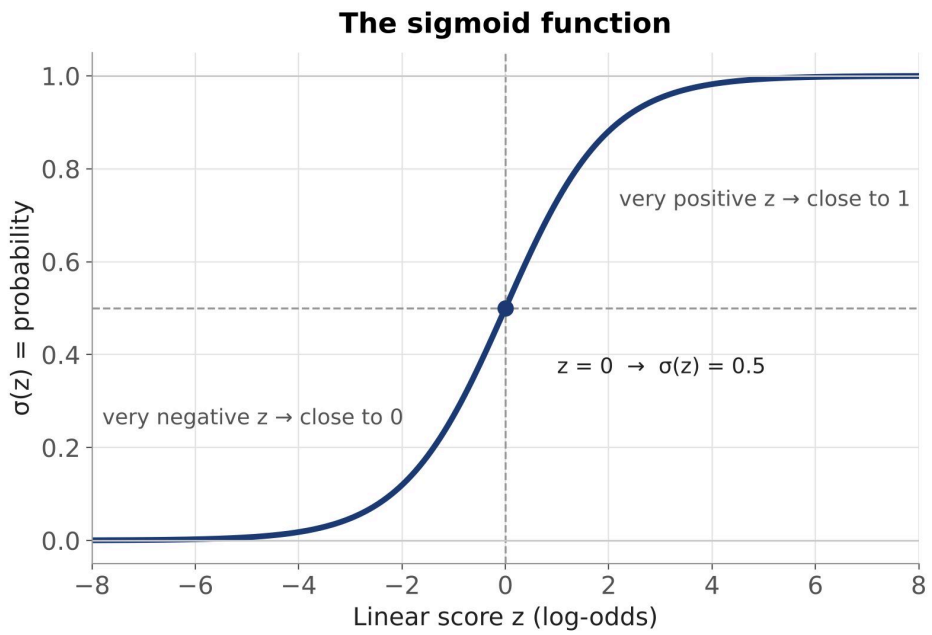
→ close to 0

Zero

→ 0.5

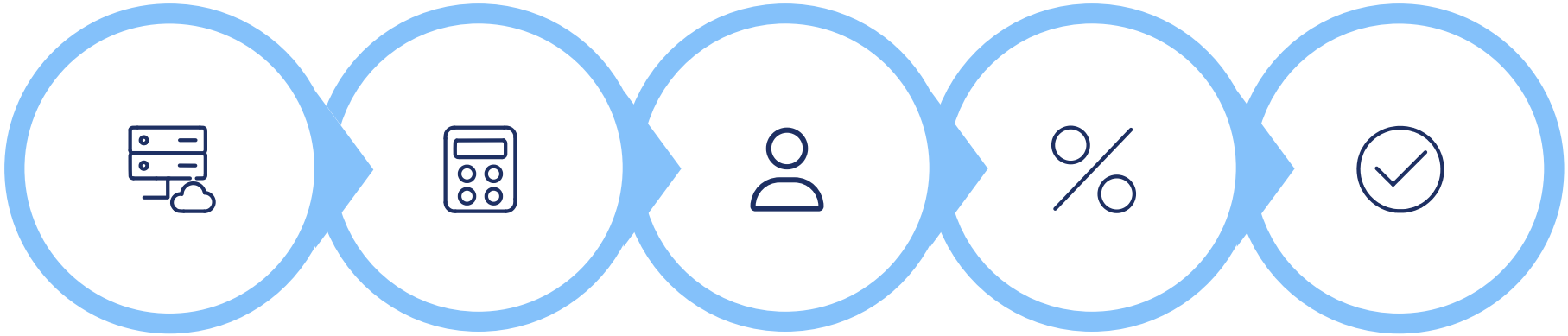
Very positive

→ close to 1



From Features to a Prediction

Logistic Regression



Features → Linear score → Sigmoid → Probability → Class

For example:

$$P(\text{Pass}) = 0.82$$

With a threshold of 0.5: **Prediction → Pass**

Probability ≠ Class

The model produces a **probability**. The **threshold** determines the class.

Probability	Class (threshold 0.5)
0.18	Fail
0.43	Fail
0.72	Pass
0.94	Pass

The threshold can be changed. 0.5 is a common default, not a universal rule.

Logistic Regression: Decision Boundary

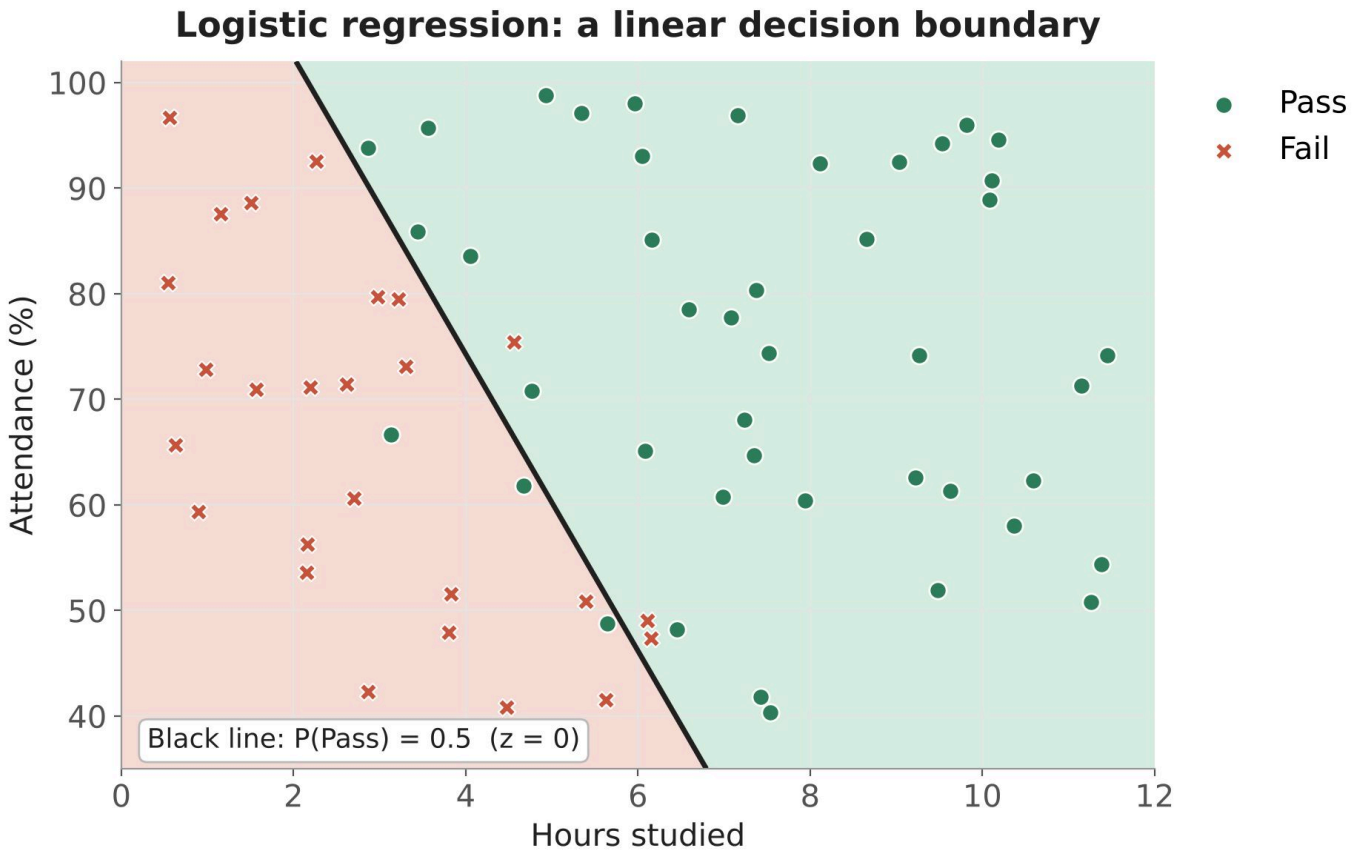
For two features:

$$z = b_0 + b_1 x_1 + b_2 x_2$$

With the default threshold of 0.5, the decision boundary is where:

$$P(y = 1) = 0.5 \quad \Leftrightarrow \quad z = 0$$

Result: a linear decision boundary in the features used.



How Does Logistic Regression Learn?

Remember Week 5: **Parameters → Predictions → Loss**

For logistic regression:



Today: Focus on the intuition. **Later:** We will look at how the parameters are actually optimised.



A Different Idea: K-Nearest Neighbours

What if we don't learn a mathematical boundary at all?

KNN: Similar observations tend to have similar classes.

For a new observation:

1. **Find the nearest training observations.**
2. **Look at their classes.**
3. **Take the majority vote.**

KNN in Action

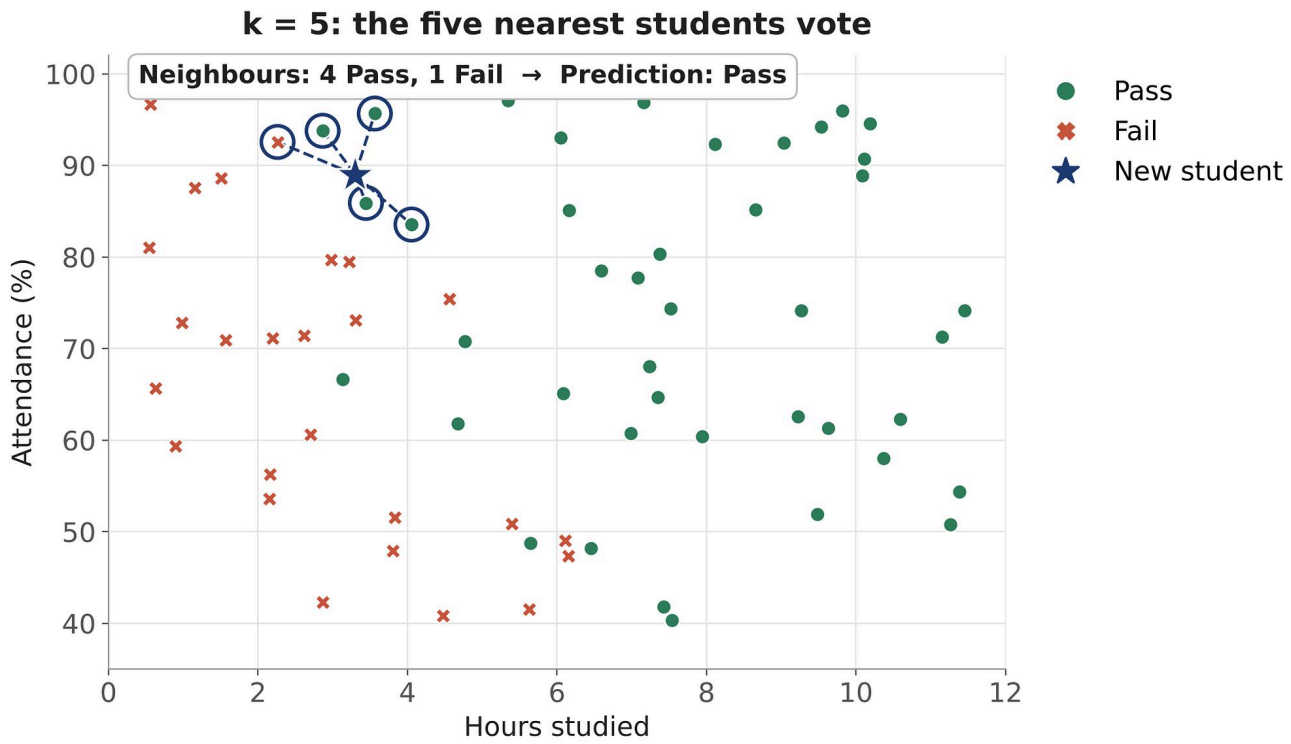
Suppose $k = 5$.

Among the 5 nearest students:

4 → Pass

1 → Fail

Prediction → Pass



What Does "Nearest" Mean?

KNN needs a way to measure distance.

For two features (Euclidean distance):

$$d = \sqrt{(x_1 - x_1')^2 + (x_2 - x_2')^2}$$

Important: The scale of the features matters. A feature measured in large units dominates the distance.

⚠ Feature scaling is especially important for KNN.

What Does k Do?

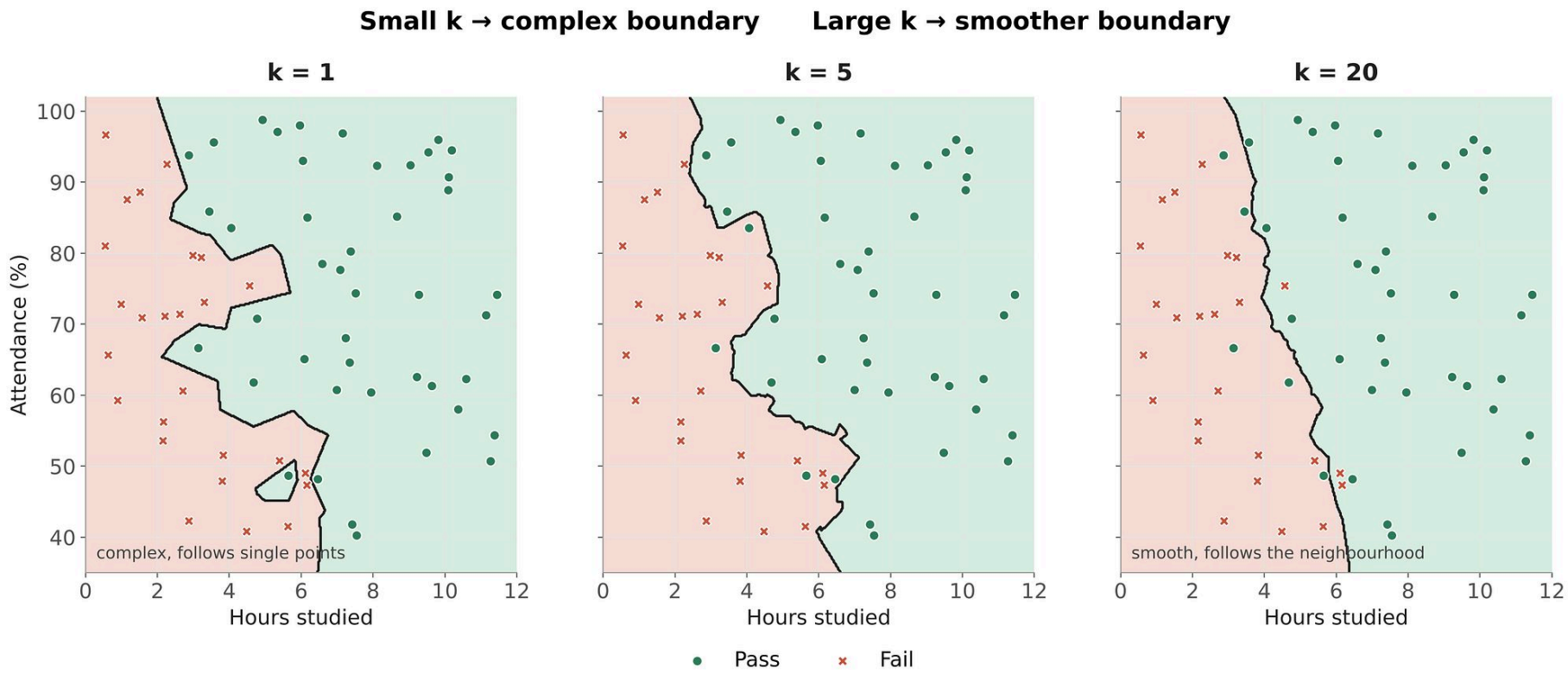
Small k

More sensitive to local patterns → more complex boundary

Large k

More influenced by the overall neighbourhood → smoother boundary

The choice of k affects model complexity.



Logistic Regression vs KNN

	Logistic Regression	KNN
Main idea	Learn a boundary	Look at neighbours
Boundary	Linear (in the features used)	Can be nonlinear
Distance-based?	No	Yes
Scaling important?	Matters with regularisation; helps optimisation	Yes, essential
Main parameter	Regularisation strength	k

Same problem. Different assumptions.

Two Ways to Classify

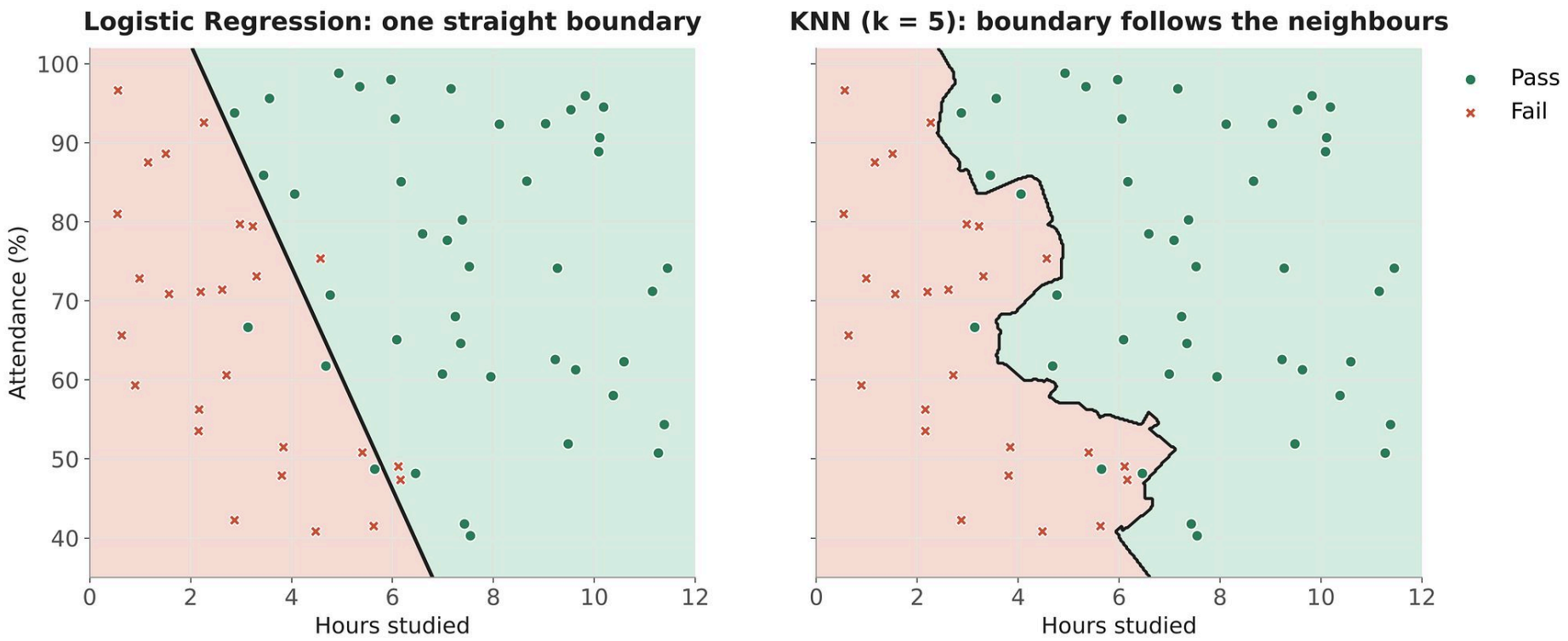
Logistic Regression

Learn a mathematical relationship → probability → threshold → class

KNN

Find similar observations → neighbours → majority vote → class

Same data, different assumptions



Hands-On: Build Two Classifiers

Using the same dataset:

01

Define X and y

02

Split into training and test data

03

Scale the features (fit the scaler on training data only)

04

Train Logistic Regression

05

Train KNN

06

Compare test accuracy

07

Change k: what changes?

Quick Check

Which statement is correct?

A.

Logistic regression outputs a category without computing a probability.

B.

KNN learns coefficients during training.

C.

Logistic regression can produce a probability for the positive class.

D.

KNN does not depend on feature scale.

Think → Discuss → Answer

Answer: ?

Key Takeaways

Classification

Predict a category, rather than a numerical value.

Logistic Regression

Linear score (log-odds) → probability → threshold → class

KNN

Nearest observations → majority vote → class

Important

Different algorithms make different assumptions about the data.

Next Time

Decision Trees

What if classification can be expressed as a sequence of questions?

- Study hours > 5?
- Attendance > 80%?
- Pass or Fail?

And what happens when we make our model too complex?

Next: Decision Trees & Model Complexity